Section

3

PRESTIGIOUS DISCOVERY SDN BHD

The Real-time System Specialist

Viscon Dua RTU Maintenance Manual

VISCON DUA REMOTE TERMINAL UNIT

Maintenance Manual

REVISION 3.2

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Chapter 1

Preventive Maintenance

This chapter describes the recommended preventive maintenance exercises for Viscon Dua RTU. It is divided into several sections and may be handled by different parties.

Viscon Dua Remote Terminal Unit (V2RTU) requires regular preventive maintenance exercises to ensure the system operates in its optimum performance. Preventive maintenance can prevent from any system breakdown during its operation. This exercise is also can predict any possible problem in future and propose appropriate action to prevent problem from occur.

The time frame of each maintenance exercise can be planned between every three to 12 months depending on how critical the system operation. The size of the system in terms of number of hardware and peripherals and point counts is also to be considered when planning the frequency of the maintenance exercise.

Maintenance Work Information

For a good practice, the maintenance team is advised to retain the following information every time they carry out the system maintenance work:

- Personnel particulars who carry out maintenance work and his/her endorsement
- Personnel particulars who witness the maintenance work and his/her endorsement
- Date the maintenance work is carried out
- Time start and finish work

Scope of Maintenance

Preventive maintenance for V2RTU can be divided into three scopes which are:-

- Power supply/ battery charger
- Communication equipment/ modem
- Remote Terminal Unit

Power Supply/ Battery Charger

There are three items to be inspected within RTU power supply and battery charger which are:

- **Physical inspection** – battery charger terminals are clean and in good condition
- **Backup supply –** voltage for each backup supply is ±10% from nominal voltage
- **Voltage Output at RTU DC MCB** – voltage output for DC MCB is ±10% from nominal voltage (nominal value is 110VDC or 48VDC)
- **Voltage Output at RTU AC MCB** – voltage output for AC MCB is ±10% from nominal voltage (nominal value is 240VAC)

Communication Equipment/ Modem

There are four items to be inspected within communication equipment/modem which are:

- **Physical inspection** – communication equipment/modem is in good condition and clean
- **Wiring connection –** all wirings are properly connected and labeled
- **Communication parameters and setting** – all parameters and setting are configured correctly
- **Communication/ Modem functionality** – Communication equipment/ modem is functioned correctly

To check the functionality of leased line modem:

1. Observe status at the display. It should display 'Connect 9' to show the modem is correctly connected to the other pair.
2. Check the presence of the communication line.

To check the functionality of Ethernet-RS232 convertor:

1. At Ethernet-RS232 convertor, replace connection to MS with PDU.
2. From PDU, send command to Poll RTU. PDU should be able to communicate with the RTU.
3. Remove connection to PDU and replace with connection to MS. It should operate normally.

Remote Terminal Unit

There are four items to be inspected within RTU panel which are;

- **Physical inspection** – inspect the condition of the RTU panel to ensure the panel is free from dirt and in good condition
- **Wiring connection** – inspect the connection of the wiring within the RTU panel to ensure all connections are terminated correctly and properly labeled
- **RTU parameters and setting** – check the RTU parameters and setting to ensure the configurations are correct according to the RTU As-built Drawing and Master Sheet
- **RTU functionality** – randomly check the RTU functionality for Digital Input, Analog Input and Digital Output

To check the RTU Parameters:

1. Connect PDU with RTU to respective I/O module at DIAGNOSTIC port.

2. Ensure PDU is working in 'NORMAL' mode to avoid any unintended editing on configuration or I/O setting information.

3. On Configurator window of PDU, select *System Setting*. Go to *Parameter* tap.

4. PDU will display the current RTU parameters. Check the parameters information to ensure no unintended changes is made.

To check the RTU I/O setting:

1. Connect PDU with RTU to respective I/O module at DIAGNOSTIC port.

2. Ensure PDU is working in 'NORMAL' mode to avoid any unintended editing on configuration or I/O setting information.

3. On Configurator window of PDU, select *I/O Setting* command. Get I/O configuration according to card installed.

4. Compare the result with Master Sheet.

To check the functionality of Analog Input point:

1. Isolate all RTU I/O point terminal blocks from equipment at site.

2. Connect PDU to RTU I/O module at MASTER port, replacing connection to Master System.

3. By using current injector, inject current at respective terminal block and observe reading at PDU Translator window.

4. The reading that shown on PDU should tally with the current injected at terminal block.

To check the functionality of Digital Input point:

1. Isolate all RTU I/O point terminal blocks from equipment at site.

2. Connect PDU to RTU I/O module at MASTER port, replacing connection to Master System.

3. Short any positive common terminal (terminal block 37 to 40 of DI module) with any Digital Input point terminal block on the same module (terminal block 1 to 32 of each DI card).

4. Observe reading at PDU Translator window. The value shown on PDU should change from '0' to '1' when the respective point is shorted.

To check the functionality of Digital Input point:

1. Isolate all RTU I/O point terminal blocks from equipment at site.
2. Connect PDU to RTU I/O module at MASTER port, replacing connection to Master System.
3. Set multi-meter to check continuity.
4. Prop negative point terminal of the multi-meter to the first pin of DO point and positive point terminal to the second pin of DO point.
5. On PDU, send a control command to the respective DO point. Multi-meter will show reading to indicate that the control command has been successfully executed.

Chapter 2

Card Testing

This chapter describes Viscon Dua RTU card testing that involve testing at the individual card level for CPU card, Digital Input card, Digital Output card and Analog Input card.

There are some cases where the person who carries the maintenance work at site will bring back individual card and suspect only that particular card is not function properly. The following information shows how individual card is tested to make sure it has no problem and can be installed back to the system to make it running as usual.

CPU Card Testing Procedures

The **CPU card** testing procedures involve:-

- DIN64 connector, Master and Diagnostic tests
- Functional test
- Power supply test
- Link port test

DIN64 Connector, Master and Diagnostic Tests

1. Insert DOM into J5 connector on the CPU Card. Pre-program the DOM to have a baudrate of 9600 bps, Station and RTU addresses are both equal to 1 respectively and Port number is set to 0.

2. Insert Mity-SOC into J2 connector on the CPU card. Ensure the Mity-SOC is inserted correctly.

3. Connect a CPU card to the CPU CARD slot on the backplane, a DO card to I/O CARD 1 slot, a DI Card to I/O CARD 2 slot, and an AI card to I/O

CARD 3 slot. Make sure DO card DIP switch is set to 1, DI card is set to 2, and AI card is set to 3.

4. Connect RS232 cable from the CPU Diagnostic port to PC that runs PDU.

5. Turn ON the power supply. Ensure CPU Run LED blinks. This shows that the CPU card is initially OK.

6. On PDU, click on the **Port status** tab and click on **Get** button. This is to get the station and RTU numbers. If there is no reply from the CPU, click on the Communication menu, and then choose **Auto seek**.

7. On the PDU, click on the **General Parameter** tab. Click on **Get** button. RTU should respond with the corresponding card types as shown on the Serial numbers.

8. Turn OFF the power supply. Change the DO DIP switch to 2 and DI DIP switch to 1. Turn ON the power supply.

9. On PDU, click on the **RTU port config**. Tab. In Reset column select **Configuration**. RTU will reboot. Repeat step 7 above.

10. Change the cable from diagnostic to Master port. On PDU, click on **Auto poll**. The PDU should receive replies from RTU as in the Receive window.

Functional Test

1. For DI point test, please refer to Test Procedures for DI card. Test only point 1 and 8.

2. For DO point test, please refer to Test Procedures for DO card. Test only point 1 and 8.

3. For AI point test, please refer to Test Procedures for AI Card. Test only point 1 and 8.

Power Supply Test

1. Measure the power supply to the CPU. Ensure that the input voltage is almost similar to power supply voltage. Vin = ____

2. Measure the voltage at test points on the CPU card, +5 and + 12 V. Reading on the CPY card for + 5 V = _____ and +12 V = ____.

Link Port Test

1. Power up the test CPU and another to be as slave CPU. This CPU is set to Port 1, RTU and Station addresses are 1, and baudrates for both master and diagnostic ports are set to 9600bps.

2. Connect a twisted RS422 cable from test CPU to slave CPU.

3. On PDU, click on **RTU port status**. Click **Get** button, ensure that port 0 button is active green. Click on Port 1, the text Active yellow will appear, then click **Set** button.

4. Observe the inter-port polling for 2 minutes. If the Link LED's at the test CPU blink, it shows that the CPU is good.

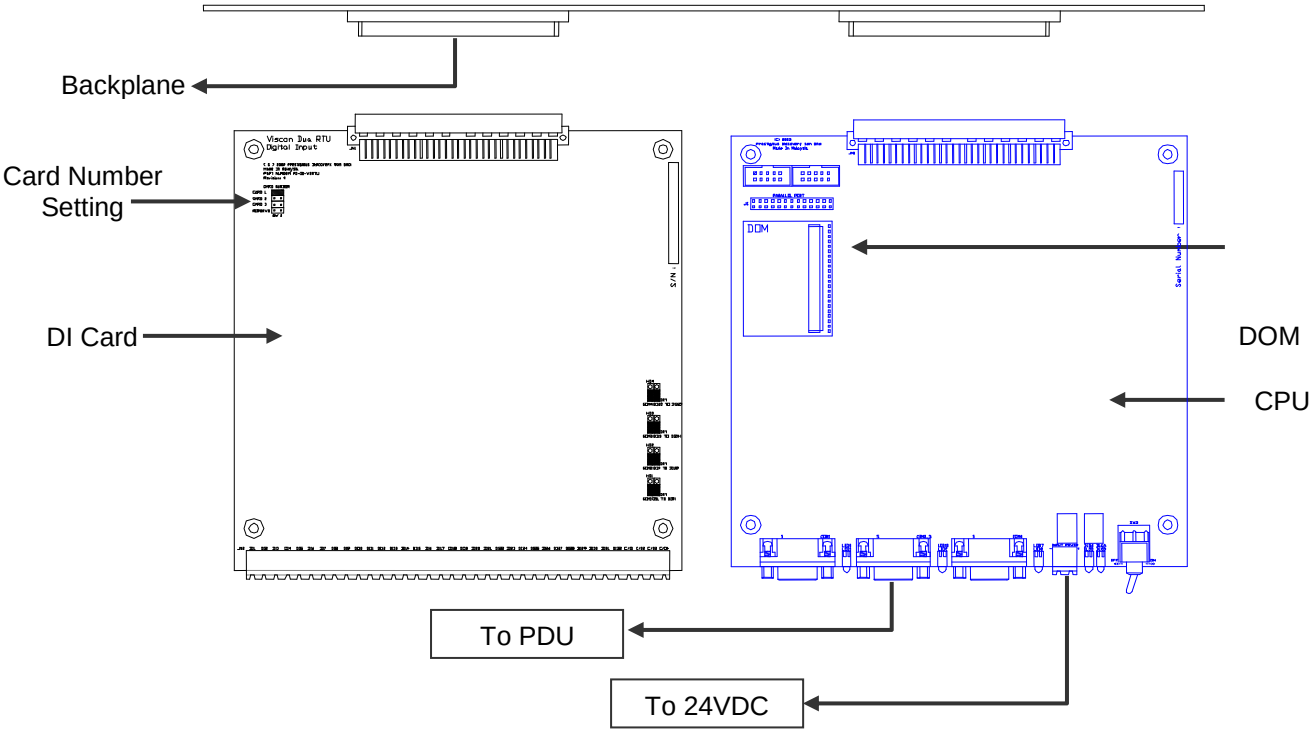


Figure 2-1: RTU configuration with CPU card and DI card for Card number 1.

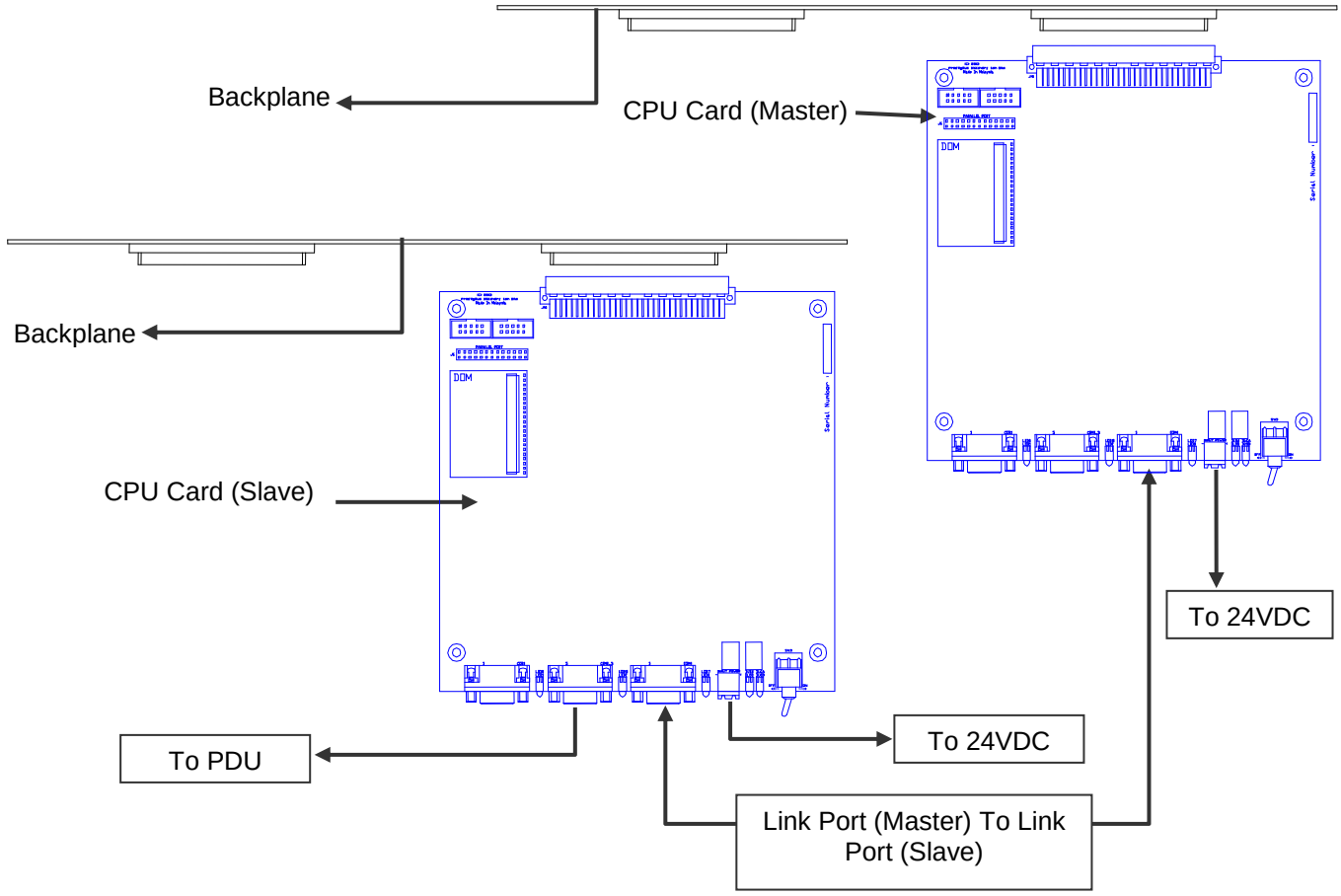


Figure 2-2: RTU Master and Slave configuration for Link port test

DI Card Testing Procedures

DIP switch test

1. Set the DI DIP switch-card number to 1.
2. Connect the DI card and a CPU card to a backplane. The CPU is pre-configured to have 9600 bps for Master and Diagnostic comm. ports, RTU number and Station number 1 and Port number 0.
3. Turn ON the power supply. Check if CPU Run LED blinks. This shows that the DI card is initially OK.
4. On PDU, click the **General Parameter** tab command. Click on **GET** button. Check information on Serial numbers. CARD 1 should be available, CARD 2 and CARD 3 are not available.
5. Turn OFF the power supply. Change the DI DIP switch-card number to 2.Turn ON the power supply.
6. On PDU, click on the **RTU port config**. On the Reset box, choose the **Configuration**. Click **Set** button. RTU will reboot.
7. Repeat step 4. Check information on Serial numbers. CARD 2 should be available, CARD 1 and CARD 3 are not available.
8. Repeat steps 5 to 7 for DI switch-card number set to 3. Observe the result.

DI Point Test

1. Set the DIP switch-card number to 1.
2. Connect a 'loop' 4-pin connector to point 33 to 36. Connect the loop to -24VDC. This is to group the common points.

3. Power ON the RTU. Ensure the CPU RUN led is blinking. On PDU, click on Auto poll.

4. Short +24VDC to DI point 1. Observe the PDU. On the V2PDU: Packet Translator window, point 1 should give a reading as Typeid 1: Digital Input (Single Point).Please make sure that:

    - the data displayed is tally with the exercise done on the point. If +24VDC is shorted to point 1, the RTU will report that point 1 has a value of '1' and if open, the value is '0'.

    - the point number displayed is tally with the DI point of test.

5. Repeat the step 4 for points 2 to 32.

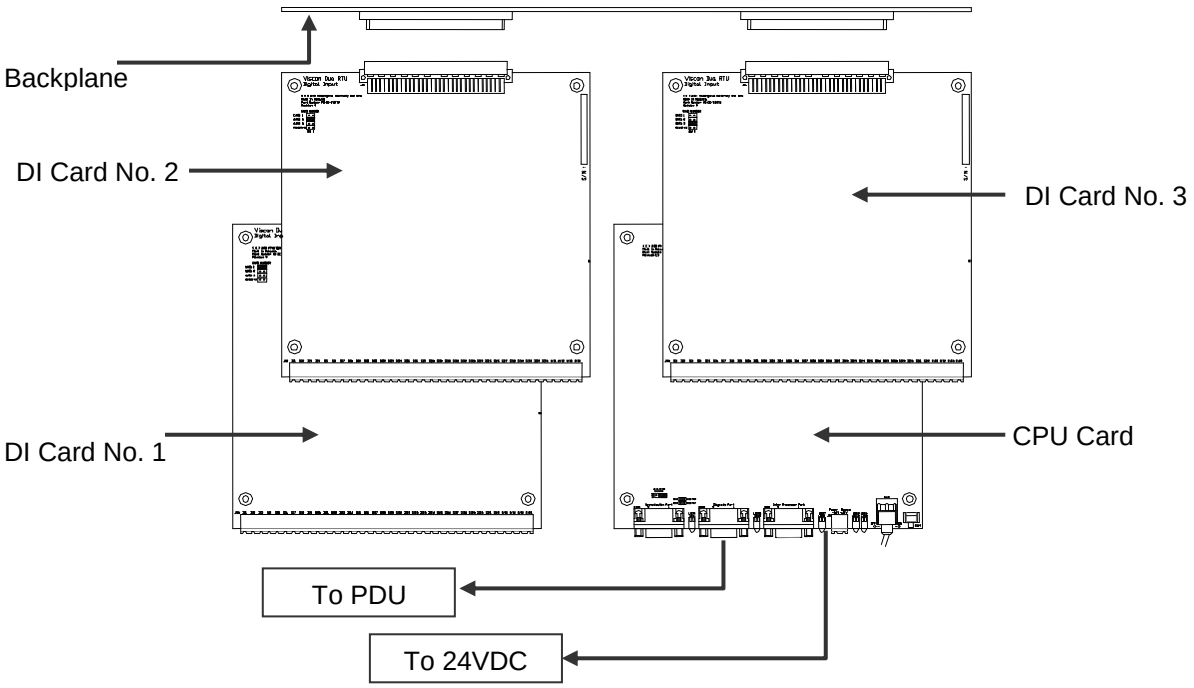


Figure 2-3: RTU configuration with 3 DI cards

DO Card Testing Procedures

DIP switch test

1. Set the DO DIP switch-card number to 1.
2. Connect the DO card and a CPU card to a backplane. The CPU is pre-configured to have 9600 bps for Master and Diagnostic comm. ports, RTU number and Station number 1 and Port number 0.
3. Turn ON the power supply. Check if CPU Run LED blinks. This shows that the DO card is initially OK.
4. On PDU, click the **General Parameter** tab command. Click on **GET** button. Check information on Serial numbers. CARD 1 should be available, CARD 2 and CARD 3 are not available.
5. Turn OFF the power supply. Change the DO DIP switch-card number to 2.Turn ON the power supply.
6. On PDU, click on the **RTU port config**. On the Reset box, choose the **Configuration**. Click **Set** button. RTU will reboot.
7. Repeat step 4. Check information on Serial numbers. CARD 2 should be available, CARD 1 and CARD 3 are not available.
8. Repeat steps 5 to 7 for DO switch-card number set to 3. Observe the result.

DO Point Test

1. Set the DO DIP switch-card number to 1.
2. Connect +24 VDC to point 35 and –24 VDC to point 33. This is to enable Control Enable Switch (CES) of the DO points.
3. Connect a control test jig to point 1 to 16.
4. Turn ON the power supply. Check if CPU Run LED blinks.

5. On PDU, click on **Test: Digital control**. Set the point from 1 to 16. Click the ON checkbox. Click Start

6. Check if LED on the control test jig lights on and off sequentially from point 1 to 16.

Observation: *Please make sure that only one LED lights on when control is executed. Otherwise, the DO card is faulty.*

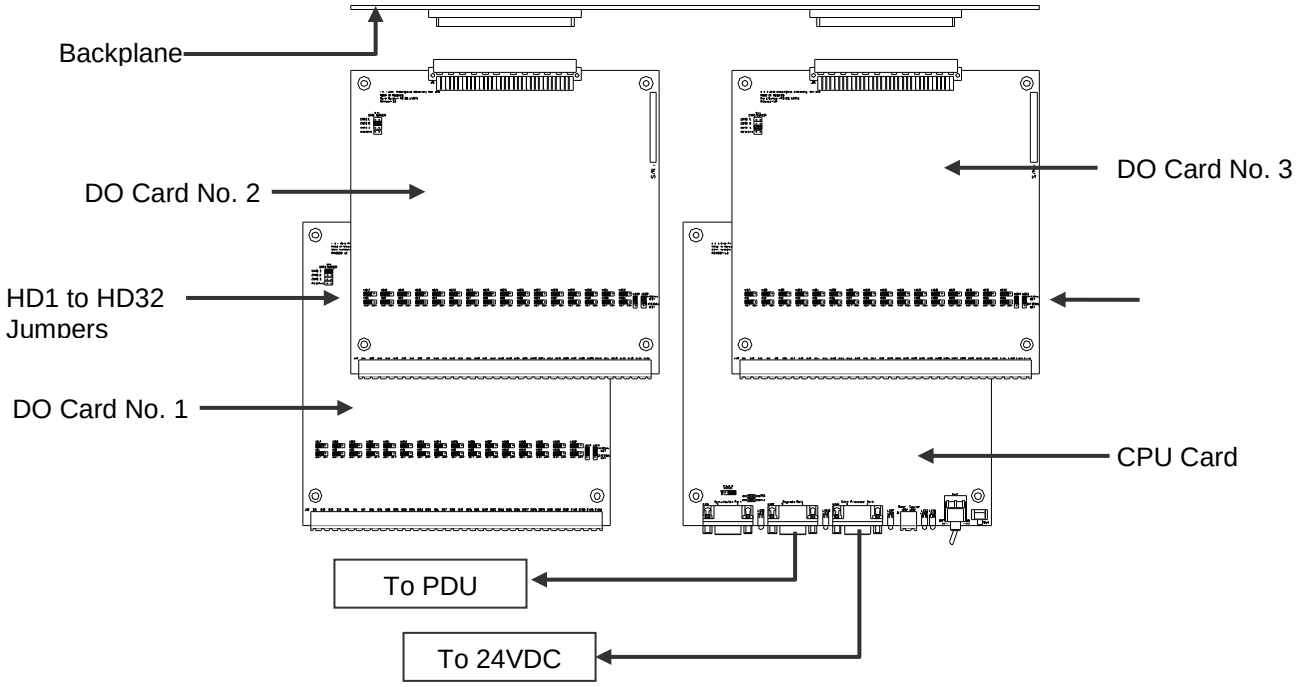


Figure 2-4: RTU configuration with 3 DO cards

AI Card Testing Procedures

DIP switch test

9. Set the AI DIP switch-card number to 1.
10. Connect the AI card and a CPU card to a backplane. The CPU is pre-configured to have 9600 bps for Master and Diagnostic comm. ports, RTU number and Station number 1 and Port number 0.
11. Turn ON the power supply. Check if CPU Run LED blinks. This shows that the AI card is initially OK.
12. On PDU, click the **General Parameter** tab command. Click on **GET** button. Check information on Serial numbers. CARD 1 should be available, CARD 2 and CARD 3 are not available.
13. Turn OFF the power supply. Change the AI DIP switch-card number to 2.Turn ON the power supply.
14. On PDU, click on the **RTU port config**. On the Reset box, choose the **Configuration**. Click **Set** button. RTU will reboot.
15. Repeat step 4. Check information on Serial numbers. CARD 2 should be available, CARD 1 and CARD 3 are not available.
16. Repeat steps 5 to 7 for AI switch-card number set to 3. Observe the result.

Point Tests – 1mA

1. On **AI card**, set card number to 1.
2. On the **AI card**, set the jumpers HD1A to HD16A to 1 mA. As shown in PCB.
3. Connect a current injector to point 1. Ensure that the +ve input is left and –ve is right point next to it. In term of physical point, connect +ve to pin 1 and –ve to pin 2.
4. Power ON the RTU. Ensure the RUN led is blinking.

5. Inject 100% of the test input value. In this case, inject 1 mA. Measure the voltage at pin 16 of IC U3, MAX197BCNI with respect to TP4 GND. The voltage should be 5 VDC.

6. Decrease the input to 50% and then 0%. The measured voltage for 50% should be half and 0% should be 0. In case of 1 mA, decrease in 50% will give 2.5 VDC and 0% will 0 VDC.

7. Repeat procedures 5 to 6 for point 1 to 7. As for the measured point at MAX197BCNI, please use this reference.

Point of test	Ux*, MAX197BCNI pin number
0	16
1	17
2	18
3	19
4	20
5	21
6	22
7	23

*Note: if point 0 to 7, x is 3, for points 8 to 15, x is 4.

8. For point 8 to 15, repeat procedures 5 to 6 but the ensure the MAXIM points measured is U4. Please refer to 7.

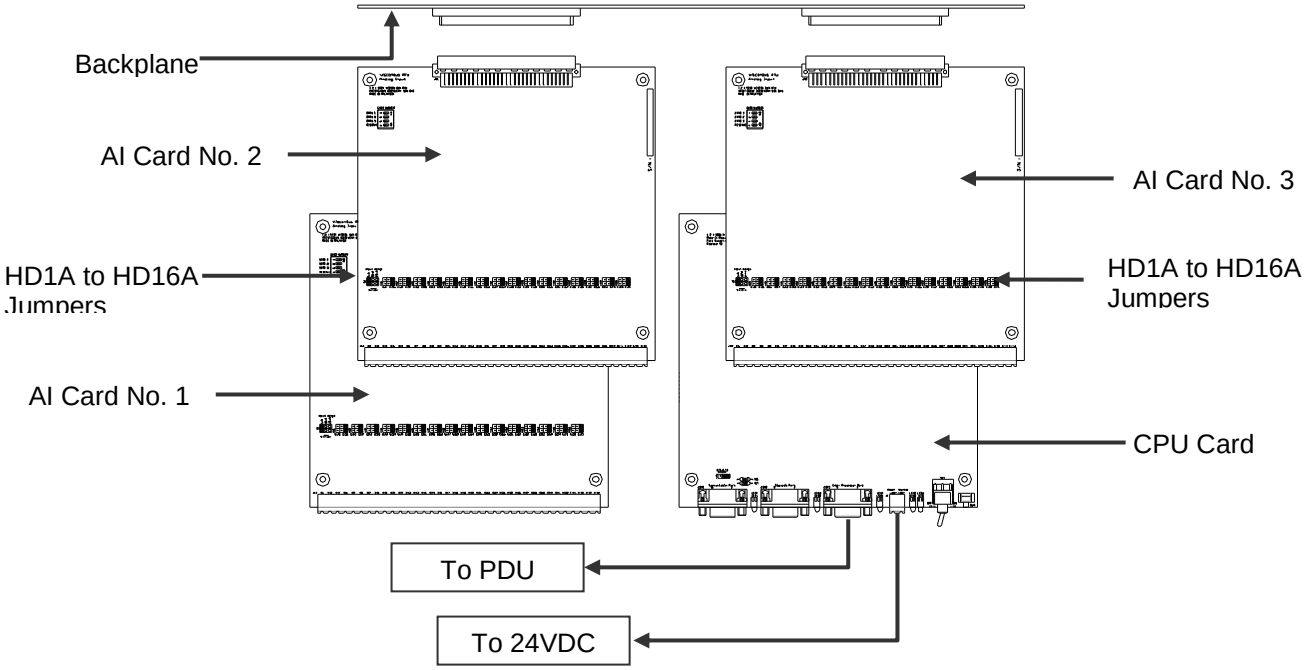


Figure 2-5: RTU configuration with 3 AI cards

RS422 Card Test Procedures

RS422 card test procedures involve:-

- General Setting – RS422 as Primary
- Power ON test
- RS422 Comport Test
- RS422 as Secondary test

General Setting- RS422 as Primary

1. Place the RS422 card on the PDIL module together with a PDM card. Connect a 2-wire cable (power cable) from PDM to CN1 on RS422 card.
2. Set H2A to H2D jumpers to PRI.
3. Set H2E and H2F jumpers to Link 7.
4. Set H1A to H1J jumpers to RS422.
5. Set H7 and H5 jumpers to ON and HIGH respectively.
6. Ensure NO JUMPERS on HBA to HBC.
7. Set H3A to H4A jumpers to LINK 7.

Power ON Test

1. Turn ON the power supply.
2. Ensure that the PPS IN LED is blinking in green color. This color indicates that no GPS is detected. If GPS is available, the color will be red.
3. Ensure that the ALARM OUT LED is not ON.

RS422 Comport Test

1. Prepare 2 sets of CPUs. One is set to Port 0 and the other is set to Port 1.
2. Connect RS422 cable from Link comport Port 0 to Link 1 RS 422 card and another RS422 cable from Link comport Port 1 to Link 2.

3. Turn ON CPUs and PDIL module. Ensure that the Link comport LED's for both CPUs are blinking.

4. Change the RS422 cable from Link 2 to Link 3 and continue to Link 6. For each change, observe the Link comport LED's and ensure that they are blinking without hiccups for at least one minute.

5. If GPS is available, ensure of its power source. If it gets from RS422 card, set the jumper according to its voltage. For example, if GPS needs +24VDC, connect the HBC jumper on RS422 card.

6. Observe the PPS IN LED. It will blink red which indicates the presence of a GPS signal.

RS422 as Secondary Test

1. Place the RS422 card on the PDIL module together with a PDM card. Connect a 2-wire cable from PDM to CN1 on RS422 card.

2. Prepare another set of PDIL module with settings set as 1.1 above.
3. Set H2A to H2D jumpers to SEC.
4. Set H2E and H2F jumpers to Link 1.
5. Set H1A to H1J jumpers to RS422.
6. Set H7 and H5 jumpers to ON and HIGH respectively
7. Ensure NO JUMPERS on HBA to HBC.
8. Set H3A to H4A jumpers to LINK 1.
9. Connect a RS422 cable from PRI PDIL LINK 6 to SEC PDIL LINK 1.
10. Observe the PPS IN LED. IT should blink red.
11. Connect RS422 cable from LINK comport Port 1 to LINK 2 of SEC PRI. Observe the LINK comports LED's.

12. Change the RS422 cable from Link 3 to Link 4 and continue to Link 6. For each change, observe the Link comport LED's and ensure that they are blinking without hiccups for at least one minute.

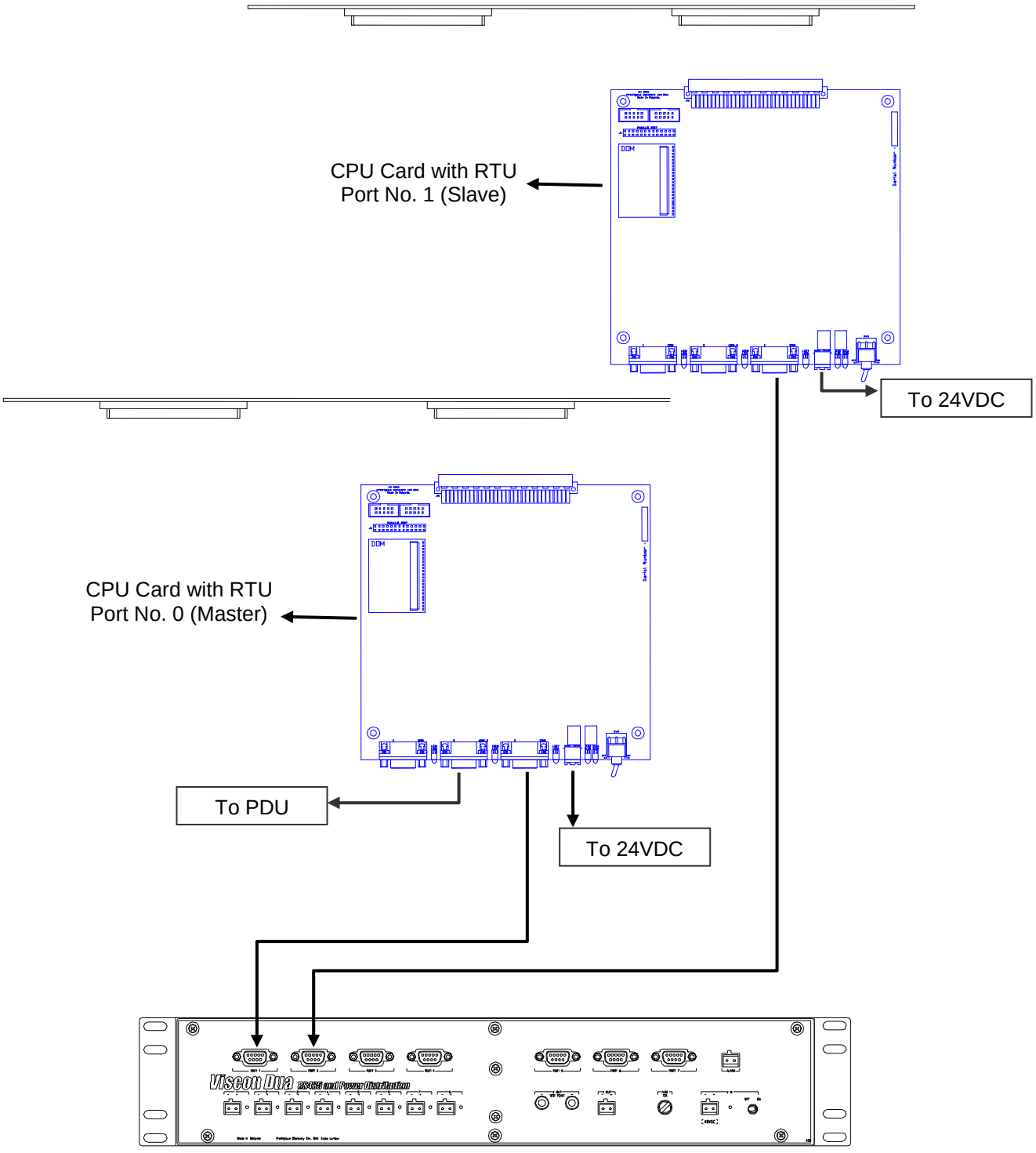


Figure 2-6: 2 RTU's connected to RS422 Port 1 and Port 2.

PDM Test Procedure

Power ON test

1. Connect 110VDC to Vin -pin connector on the PDM. Please ensure the right polarity as printed on the label.

2. Toggle the switch to ON.

3. Ensure all LED's are ON.

4. Using multi-meter, check the voltage at TEST POINT, V OUT, and 2-pin connectors from 1 till 8.

5. Record your measurement voltage at:

   TEST POINT = _____ VDC
   V OUT = _____ VDC

   1 = _____ VDC
   2 = _____ VDC
   3 = _____ VDC
   4 = _____ VDC
   5 = _____ VDC
   6 = _____ VDC
   7 = _____ VDC
   8 = _____ VDC

6. Please note that acceptable tolerance for the measurement voltage of (5) above is $\pm$ 5 VDC from the input voltage of (1).

Distributed Inter-port Link Test Procedure

Comport Test

1. Prepare 2 sets of CPUs. One is set to Port 0 and the other is set to Port 1. Port 0 is configured to poll Port 1. Prepare 1 set of PDIL module

2. Connect RS 422 cable from CPU Port 0 to RS422 card Link 0. Connect RS422 cable from RS422 card Link 6 to RS485 card Link 1. Connect RS 422 cable from CPU Port 1 to RS485 card Link 2.

3. Turn ON CPUs and PDIL module. Ensure that the Link comport LED's for both CPUs are blinking.

4. Change the RS422 cable from RS 485 card Link 2 to Link 3 and then continue to Link 6. For each change, observe the Link comport LED's for both CPUs and ensure that they are blinking without hiccups for at least one minute.

Backplane Test Procedure

1. Connect a CPU card to the CPU CARD slot on the backplane, a DO card to I/O CARD 1 slot, a DI Card to I/O CARD 2 slot, and an AI card to I/O CARD 3 slot..Make sure DO card DIP switch is set to 1, DI card is set to 2, and AI card is set to 3.

2. Pre-configured the CPU to detect IO cards as above.

3. Turn ON the power supply. Check if CPU Run LED blinks. This shows that the IO cards are initially OK.

4. On PDU, click on the **General parameters** tab. Click on **Get** button. RTU should respond with the corresponding card types as shown on the Serial numbers.

Chapter 3

RTU Troubleshooting

This chapter lists down all the known problems appear during the RTU operation and maintenance. For each problem, there are several possible reasons followed by suggestion of solution.

RTU Hardware/Software Troubleshooting For known Problems

General Problems

1. Problem Description:

Dead RTU module with no activity or light at LED indicators on CPU Card

Possible Reason	Solution
a. Not enough external DC power supply or no external DC24V supply.	a. Check the DC24V power supply unit, cable or panel and if possible, use multi-meter.
b. DC24V input is reversed.	b. Use multi-meter to double check the dc input polarity. +ve terminal must be at right (pin2) and –ve at left (pin1).
c. DC main switch located at below RTU cabinet was turned OFF.	c. Ensure main switch at ON position.
d. Module power switches were set to OFF position.	d. Ensure each module power switch at ON position.

e. Faulty CPU card:- ▪ Failed DC-DC converter module (PS1 or PS2 or both). ▪ Failed DC filter inductor (L1). ▪ Shorted capacitor or open circuit resistors or fuse resistor TH1, TH2, TH3 or TH4.	e. Use multi-meter to determine this conditions:- ▪ Measure the input (+24V) and output (+5V, +12V, -5V) voltage to/from the dc-dc converter module. ▪ Test the inductor with continuity check on multi-meter during power OFF. ▪ Follow above procedure.

2. <u>Problem Description:</u>

All LEDs or at least 3 LEDs are continuously lit during module switched ON.

Possible Reason	Solution
a. Wrong or blank DOM being used.	a. Ensure the correct firmware DOM version is being used.
b. Faulty CPU card or I/O card.	b. Confirm the I/O card condition with PDU or replace the CPU card.

3. <u>Problem Description:</u>

Error LED at master port goes ON.

Possible Reason	Solution
a. Conflict slave port setting or wrong configuration via PDU software.	a. At PDU, make sure correct port status is obtained under Port Status Command.
b. One of slave port/module is fail or dead.	b. Remove inter-port cable for that failed module and test with the diagnostic software.
c. Link/Comms failure between master port/module and one of slave port/module.	c. Check or replace RS422 inter-port cable.
d. Faulty master module CPU board, I/O cards or DOM unit.	d. Isolate master IO card (via DIP switch) and test or replace the individual CPU board or the I/O board. If possible, replace the DOM unit.
e. Faulty any slave module CPU board, I/O cards or DOM unit.	e. Isolate slave IO card (via Card Type DIP switch – SW2) and test or replace the individual CPU board. If possible, replace the DOM unit.

f. Improper card sliding to backplane board of casing module.	f. Pull off and put in back the card into the module backplane board.

4. Problem Description:

Inter-port LED indicators (Tx&Rx) goes OFF.

Possible Reason	Solution
a. Conflict slave port setting or wrong configuration via PDU software.	a. Ensure module port no. setting and configuration using PDU.
b. Faulty inter-port cable.	b. Replace the cable.
c. Inter-port cables were plugged off.	c. Ensure the connector are properly inserted and screwed.
d. CPU card failed:- ▪ DOM Unit. ▪ Faulty LEDs (Led7 & Led8).	d. Confirm with PDU first or:- ▪ replace DOM Unit ▪ replace the CPU card.

5. Problem Description:

Inter-port LED indicator, Tx or Rx only goes OFF.

Possible Reason	Solution
a. Wrong configuration via PDU software.	a. Ensure module port no. setting and configuration using PDU.
b. Faulty RS422 inter-port cable.	b. Ensure the cable connector are properly inserted and screwed or replace the cable.
c. CPU card failed:- ▪ Faulty LEDs (Led7 & Led8).	c. Confirm with PDU first or:- ▪ Replace the CPU card.

6. Problem Description:

Link failure between RTU and PDU/VMS with no Tx Rx LED activities on Master Port CPU card for master comm. port or diagnostic comm. port.

Possible Reason	Solution
a. Wrong configuration via PDU or VMS software.	a. Ensure the configuration or setting via PDU such as RTU address number and Link number, port number, comm. baud-rate or master & diagnostic comm. port parameters and at Viscon Master System (VMS) as well. If possible, try restart the PDU software or Viscon Communication Server at VMS side.

b. Faulty RS-232 cable or communication device (eg: modem).	b. Ensure the connector are properly inserted and screwed or replace the cable. Check the communication device.
c. Master Port CPU card failed:- ▪ Faulty DOM Unit. ▪ Faulty LEDs (Led1, Led2, Led9 or Led10)	c. Confirm with PDU first or:- ▪ Replace the DOM Unit. ▪ Replace the CPU card.

I/O Point Operation Related Problems

7. Problem Description:

No status changes are obtained via Digital Input Card.

Possible Reason	Solution
a. Wrong configuration/setting entry from VMS, PDU or wrong PDU version.	a. Double-check the PDU version (To check what PDU version that currently being used, go to About and select PDU version), setting with PDU software such as Port No., DI point numbers count, point type setting, Chattering Count & Time etc.
b. Reverse polarity connection.	b. Confirm the polarity with test meter.
c. Incorrect input voltage level to digital status input.	c. Make sure voltage level range to digital input point is between DC24V to 110V for status change.
d. Incorrect hardware DI point jumper setting or improper input common connection.	d. Ensure the point header jumpers position accordingly or common pin connection which are at pin 33, 34, 35 and 36.
e. Faulty DI point or DIO card.	e. Isolate the plant and confirm with PDU software and manually inject voltage to the faulty point.

8. Problem Description:

Unable to send Control Command from PDU/VMS via Digital Control card with no click sound from DO point relay.

Possible Reason	Solution
a. Wrong configuration/setting entry from VMS, PDU or wrong PDU version.	a. Double-check the PDU version or setting with PDU software such as Port No., DO point no. entry, point type setting (Immediate/ArmBeforeExecute), card enable/disable etc.

b. Faulty RS-232 cable or it was unplugged.	b. Check the cable with test meter for any short circuit or cut.
c. Faulty DO point or DIO card for master or slave PORT.	c. Observe carefully the point relay click sound or if possible, replace the DIO card with other known good DIO card.
d. Faulty Master or slave PORT smart-block unit.	d. Confirm with PDU and observe the Translate Window messages.

9. Problem Description:

Control Command was successfully sent with point relay clicking sound but nothing happen.

Possible Reason	Solution
a. Wrong configuration/setting entry from PDU or VMS. software.	a. Double-check the setting with PDU software such as Port No., DO point no. entry, control type setting (Off/Open, On/Close etc.), Control Time-out delay, Control Pulse delay etc.
b. Incorrect header jumper setting for that respective point.	b. Make sure header jumper at correct position either Wet or Dry.
c. Improper common wire connected.	c. Ensure the common wire condition at point 33, 34, 35 and 36 of DIO card.
d. Faulty respective DO point.	d. Connect the test meter probes to the respective DO point with continuity range and send control command from PDU. Observe the measured resistance for that point, if possible, replace the DIO card.

10. Problem Description:

No analog readings are obtained via Analog Input Cards.

Possible Reason	Solution
a. Wrong configuration/setting entry from VMS, PDU or wrong PDU version.	a. Ensure the correct PDU version or setting with PDU software such as, RTU address, Port No., AI point no. entry, analog type setting (4-20mA, configurable etc.) etc.
b. No available input current from analog transducer or current injector to AI point.	b. Use test meter with current range to measure the output current from transducer or current injector.
c. Reverse polarity connection.	c. Double check the cable with test meter.

d. Incorrect header jumper setting for that respective point.	d. Ensure the AI point header jumper position same as in point type setting.
e. Faulty respective AI point.	e. Use test meter to check AI point related electronic component such as Ln, LnA, RAn, INAn op-amp etc

I/O Point Configuration Related Problems

11. Problem Description:
Unable to change DI & DO point type and parameter setting from PDU software.

Possible Reason	Solution
a. Wrong version of PDU software.	a. Try above procedure to ensure the PDU version (I/O point operation 1(a)). Refer to manual which version to be used for the hardware version.
b. PDU software not in PDU Program Mode.	b. Ensure PDU software in Program Mode by selecting Option->PDU In Program Mode menu.
c. Wrong configuration or setting from PDU.	c. Ensure the correct software selection/option such as Port No, Point No etc before pressing SET command button.
d. Link failure between PDU and RTU.	d. Confirm the RS-232 cable condition, connector position etc.

12. Problem Description:
Unable to change or calibrate AIO point setting or parameter from PDU software.

Possible Reason	Solution
a. Wrong version of PDU software.	a. Try above procedure to ensure the PDU version (I/O point operation 1(a)). Refer to manual which version to be used for the hardware version.
b. PDU software not in Program Mode.	b. Ensure PDU software in Program Mode by selecting Option->PDU In Program Mode menu.
c. Wrong configuration or setting from PDU.	c. Ensure the correct software selection/option before pressing SET command button.
d. Incorrect header jumper position for that respective point.	d. Ensure the correct position of header jumper for that respective point such as 4-20mA, 1mA or

	10mA position.
e. Link failure between PDU and RTU.	e. Confirm the RS-232 cable condition, connector position etc.
f. Faulty AI point related electronic component such as op-amp, variable resistors etc.	f. Re-test several times with PDU and current injector to confirm the AI point condition, if possible, replace the I/O card.

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Chapter 4

Frequently Asked Questions (FAQ)

1. **Q: After the RTU setting has finished and starts in used and in a case of I/O point fault, is it the similar procedure we use to check the faulty point as during early stage of RTU configuration via PDU software?**

   A: Yes, it is similar. Check with PDU software first to confirm the I/O point condition then only move to hardware troubleshooting.

2. **Q: After configured the RTU, does the setting for the software and hardware change when RTU experienced a power loss?**

   A: No, it will not change due to all settings including communication port setting, I/O point type setting and analog calibration value are saved inside disc on module (DOM)

3. **Q: How do you know whether the transmitted or received data packet that contain status or reading are correct or not?**

   A: If you are experience with communication protocol, in this case IEC Protocol, then you may know whether the data is correct or not from the transmit/receive data packet using the PDU or at Master System.

4. **Q: Is the fault in hardware can be shown in software, PDU software for example?**

   A: Yes, but depends on what kind of fault. If it is I/O point fault, you can test the operation of that respective point using PDU software.

5. **Q: In a case of RTU problem, does the Control Centre (Viscon Master System) knows which port or section that makes the problems and what we should do in this case?**

   A: In this case, it also depends on what kind of fault. If the RTU is totally offline or dead then the control centre may notice this via the respective station page (RTU Not Found shown) or via Viscon Communication Server port status (RTU reply error). If the problem is on send control command, Control Centre will receive an error message if any control command was not successfully sent. In the case of AI or DI fault, Control Centre will notice and unchanged AI reading or unchanged DI status or display at the station page. After the problem section has been identified, you have to go to the RTU and diagnose the problem at RTU using PDU software.

6. **Q: How do I set the configuration of DI, DO and AI point accordingly? Please show in flow chart form.**

   A: For more detail procedure, please refer to the appropriate chapter. In brief, below are the simple procedures to follow.

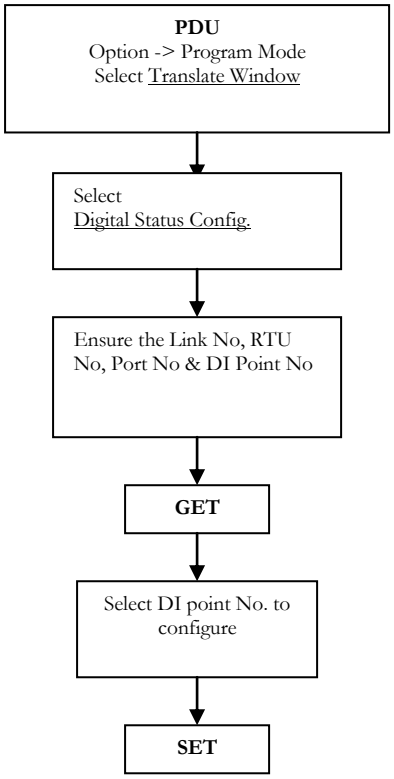


Figure 3-1: Flow diagram for setting the configuration of DI card

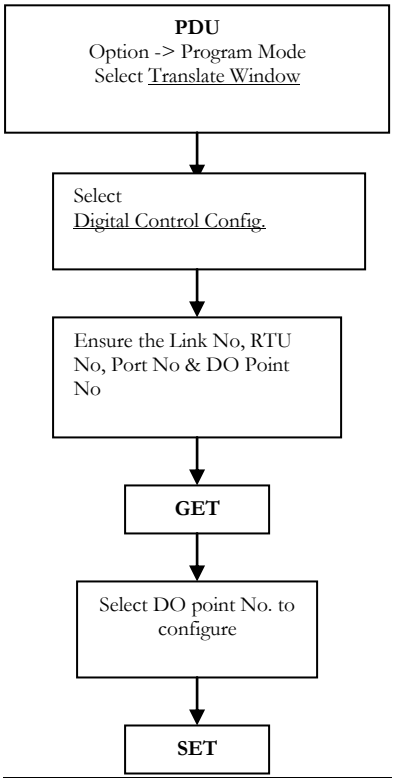


Figure 3-2: Flow diagram for setting the configuration of DO card

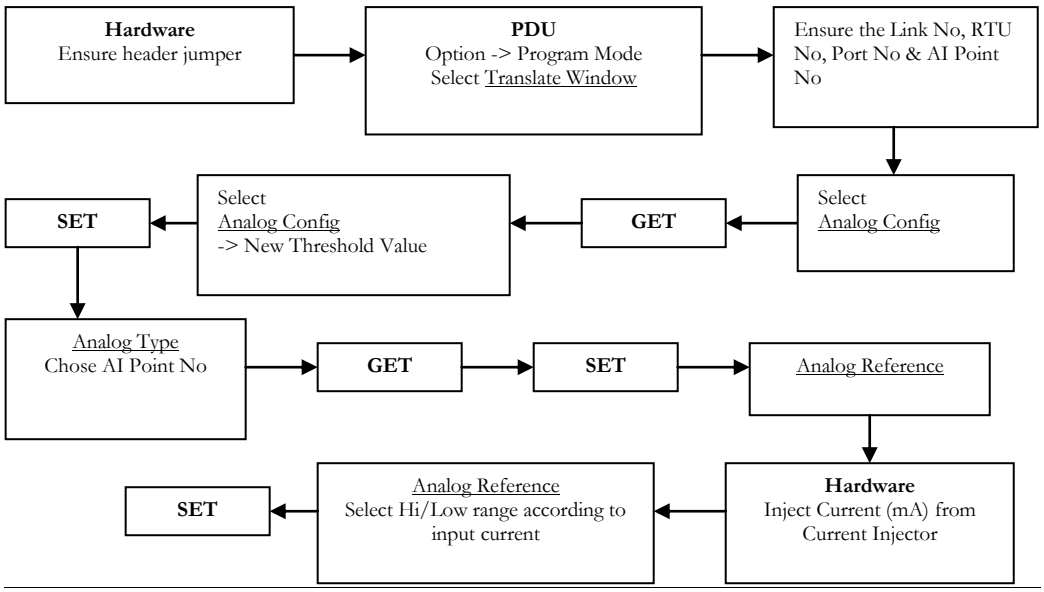


Figure 3-3: Flow diagram for setting the configuration of AI card

7. **Q: Does all the DOM is the same for different type of module, DO, DI and AI?**

   A: For DO and AI module, it is the same while for DI, it has its own version and specific for Digital Input I/O function only.

8. **Q: In a single RTU cabinet, if Port 1 was set to Link Address 0 while Port 2 & 3 was set to Link Address 3, what will happen here?**

   A: This is possible because in each RTU cabinet, it is up to the user to define what Link Address number to be used for each RTU module. The alarm will not come up even if two module are having the same port number but different Link Address number. However, in practical, it is advisable to define Link Address number and RTU Address number per cabinet to avoid system complicated.

9. **Q: Please define in detail at which port we should do the following settings, RTU Port Configuration, RTU Port Status, Time Synch, Digital Input Configuration, Digital Status, Digital Output Configuration, Digital Control, Analog Configuration, Analog Type, Analog Reference and Analog Status?**

A: Please refer below explanations:-

RTU Port Configuration → After DOM change, plug-off RS422 cable and go to each module.

RTU Port Status → You can only see which slave port is active if connection at Port 0(Master).

Time Synch → Only required at Port 0 (Master Port).

DO, DI & AI Configuration → Connect RS-232 cable to Port 0 and then selects which Port to configure.

DO, DI & AI Status → Connect RS-232 cable to Port 0, at PDU select which port to monitor then click Show RTU Module.

10. **Q: After all RTU I/O points have been set, are we required to check all usable I/O point or only some of it?**

    A: It is advisable to check all usable I/O point to avoid any wrong setting or miss-configuration and for any point problems.

11. **Q: During a card or I/O point checking and if we encounter any point fault, how we will fix it?**

    A: Please refer to Trouble-shooting manual for details.

12. **Q: If there is a problem with RTU card that currently being used, what we will do, checking the point or reconfigure the card or module?**

    A: Please refer to trouble-shooting manual for detail type of fault and the possible solution.

13. **Q: How do we connect a multiple number of RTUs at one time, for example RTU that are located far away between each other?**

A: Due to the distance factor, one of the solutions is to use a multi-drop connection between Control Centre and the multiple RTUs. Below is the connection diagram.

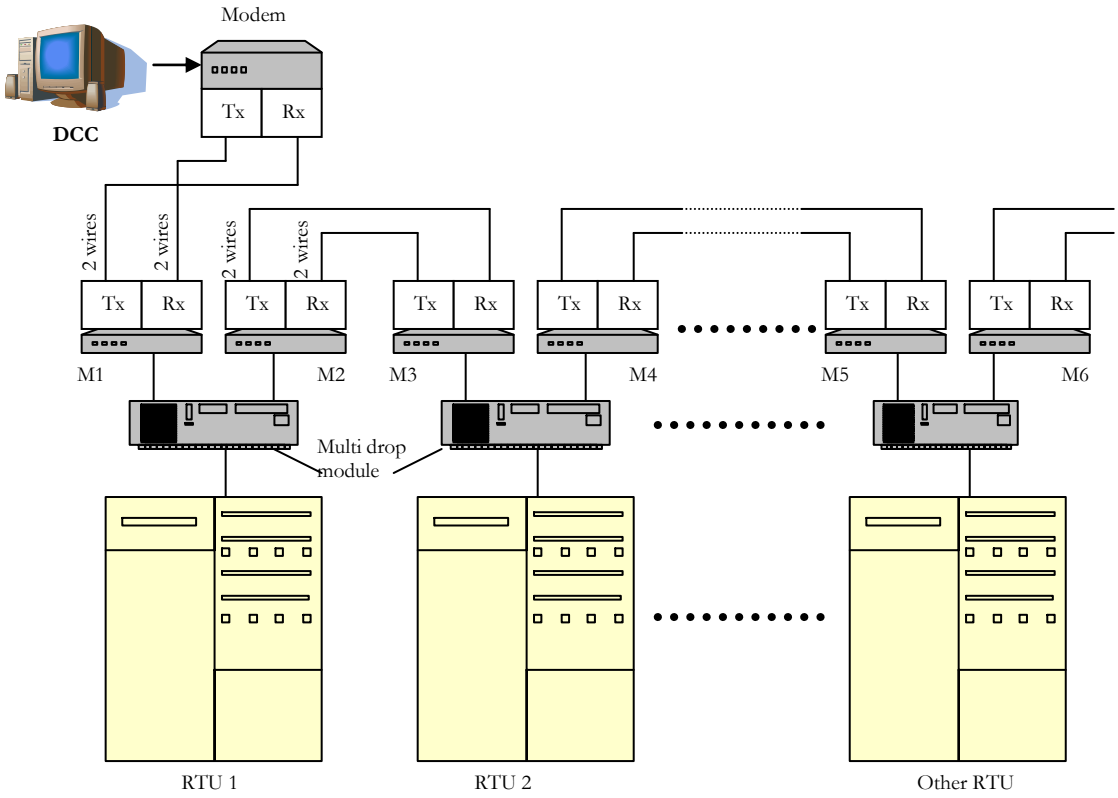


Figure 3-4: Multidrop configuration

14. **Q: What is the first step to be done during each RTU installation?**

A: Please refer to question number 6 as above.

15. **Q: Could you please tell me what are the frequently used option/selection in using PDU version 2.0b?**

A: This depending on what kind of task you are at. For example, if you are going to replace an old DOM with a new DOM, then you may follow the steps and use the PDU options that were mentioned earlier. However if you are going to do a new I/O configuration or re-configure an I/O point such like DO, DI or AI, then you may only concentrate on the I/O point status and configuration. As a

conclusion, below are listing for each category of task and its related PDU option or selection.

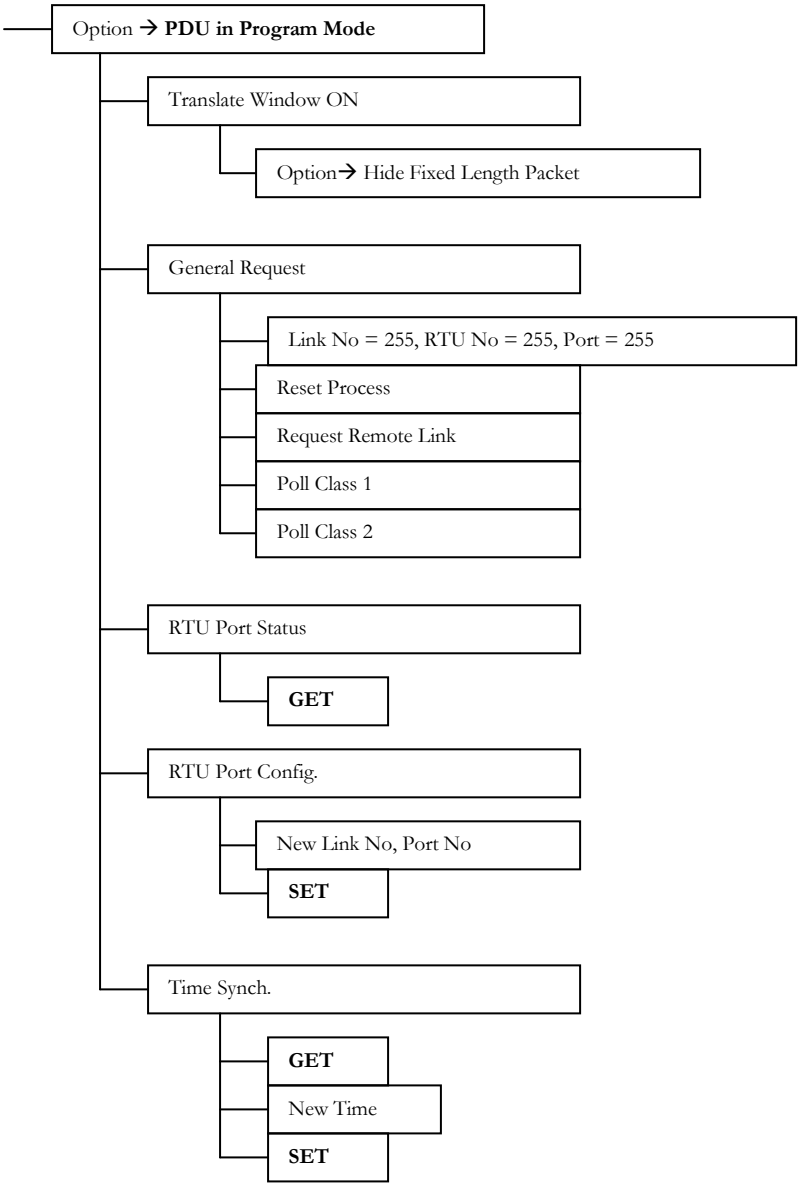
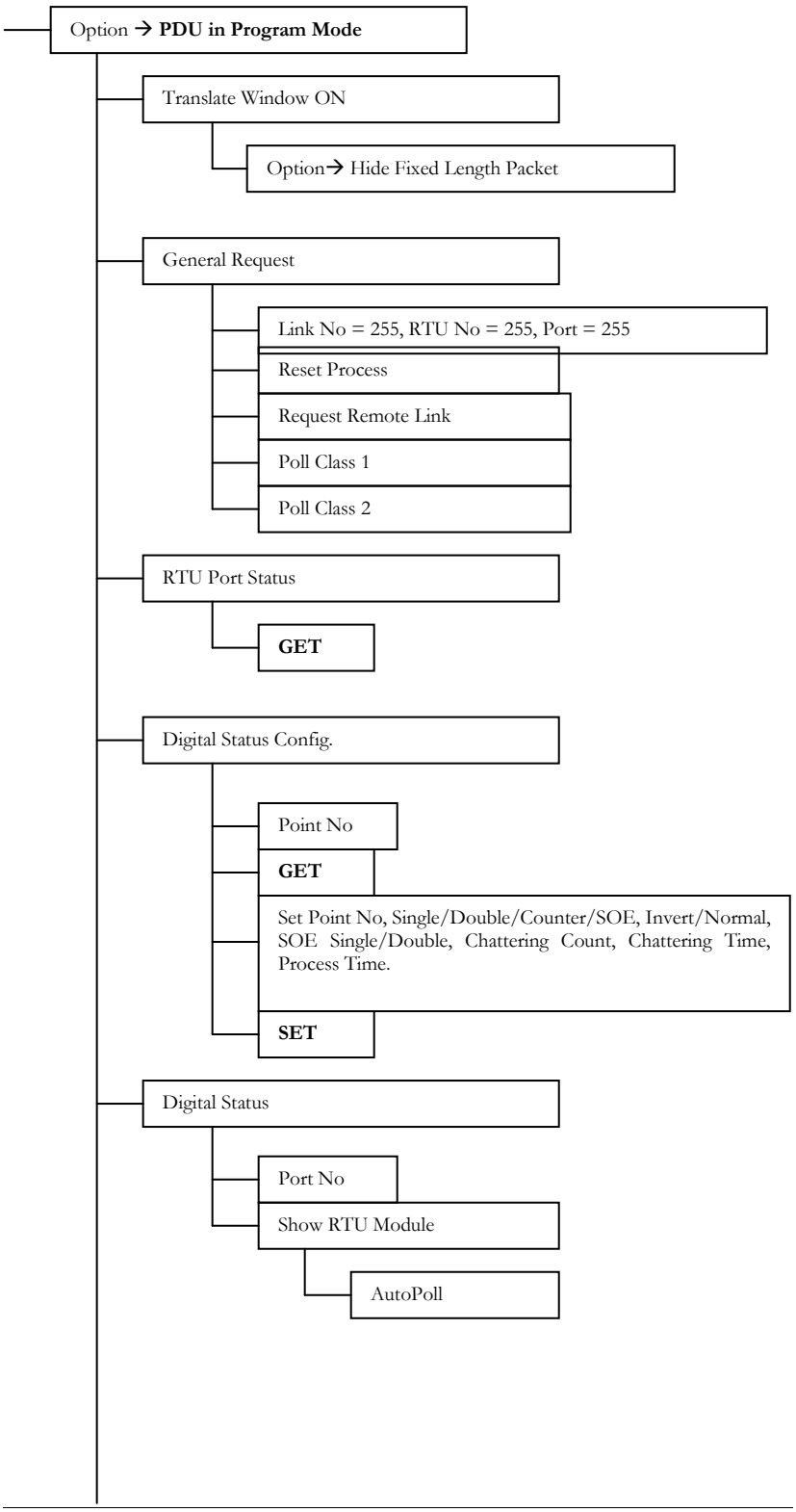
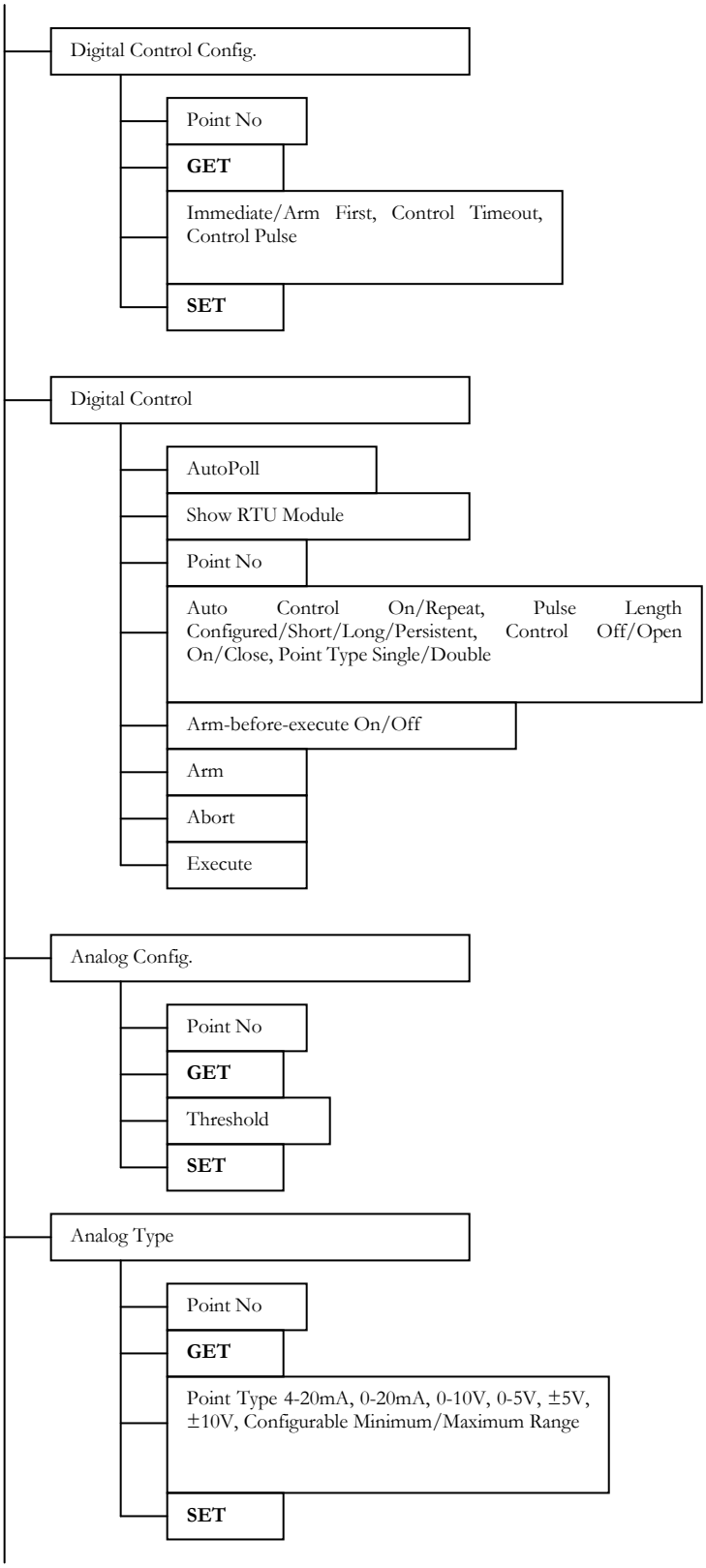


Figure 3-5: Flow diagram of PDU option and selection to replace DOM


Option → PDU in Program Mode
Translate Window ON
Option→ Hide Fixed Length Packet
General Request
Link No = 255, RTU No = 255, Port = 255
Reset Process
Request Remote Link
Poll Class 1
Poll Class 2
RTU Port Status
GET
Digital Status Config.
Point No
GET
Set Point No, Single/Double/Counter/SOE, Invert/Normal, SOE Single/Double, Chattering Count, Chattering Time, Process Time.
SET
Digital Status
Port No
Show RTU Module
AutoPoll


Digital Control Config.
Point No
GET
Immediate/Arm First, Control Timeout, Control Pulse
SET
Digital Control
AutoPoll
Show RTU Module
Point No
Auto Control On/Repeat, Pulse Length Configured/Short/Long/Persistent, Control Off/Open On/Close, Point Type Single/Double
Arm-before-execute On/Off
Arm
Abort
Execute
Analog Config.
Point No
GET
Threshold
SET
Analog Type
Point No
GET
Point Type 4-20mA, 0-20mA, 0-10V, 0-5V, ±5V, ±10V, Configurable Minimum/Maximum Range
SET

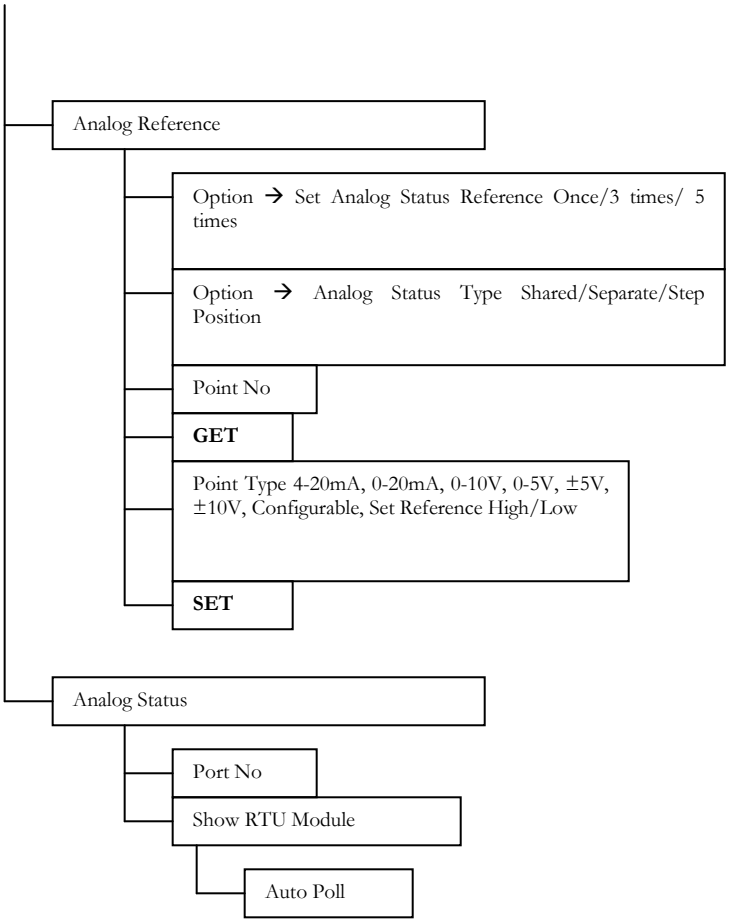


Figure 3-6: Flow diagram of PDU option and selection to change I/O Point Status & Configuration

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